

24/08/26

LAB 02

Aim: To design and implement a relational database system using DBMS concepts by creating database schemas, defining tables and relationships , populating data, and executing SQL queries for efficient data retrieval and analysis.

Q1a. Print ODIMatch table.

```
mysql> select * from ODIMatch;
```

or

```
mysql> SELECT * FROM ODIMatch;
```

MySQL	localhost:3306 ssl	matches_info_by_parth	SQL	> select * from odimatch;		
MatchID	Team1	Team2	Ground	Date	Winner	
2324	Pakistan	India	Peshwar	2006-02-06	Team1	
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2	
2357	India	England	Delhi	2006-03-28	Team1	
2377	West Indies	India	Kingston	2006-05-18	Team2	
2404	Sri Lanka	India	Colombo	2006-08-16	Abandoned	
2440	India	Australia	Mohali	2006-10-29	Team2	
2449	South Africa	India	Cape Town	2006-11-26	Team1	
2480	India	West Indies	Nagpur	2007-01-21	Team1	
2493	India	west Indies	Vadodara	2007-01-31	Team1	
2520	India	Sri Lanka	Rajkot	2007-02-11	Team2	
2611	England	India	Southampton	2007-08-21	Team1	
2632	India	Australia	Mumbai	2007-10-17	Team1	
2643	India	Pakistan	Guwahati	2007-11-05	Team1	
2675	Australia	India	Melbourne	2008-02-10	Team2	
2681	India	Sri Lanka	Adelaide	2008-02-19	Team1	
2688	Australia	India	Sydney	2008-03-02	Team2	
2689	Australia	India	Brisbane	2008-03-04	Team2	
2717	Pakistan	India	Karachi	2008-06-26	Team2	
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	

20 rows in set (0.0012 sec)

Q1b. Print Player table.

mysql> select * from Player;

MySQL localhost:3306 ssl matches_info_by_parth SQL > select * from player;

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
21001	Sehwag	Virender	India	1978	Delhi	2001
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
24001	Symonds	Andrew	Australia	1975	Birmingham	2004
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007
89001	Tendulkar	Sachin	India	1973	Mumbai	1989
90001	Lara	Brain	West Indies	1969	Santa Cruz	1990
90002	Kumble	Anil	India	1970	Bangalore	1990
91001	Jayasuriya	Sanath	Sri Lanka	1969	Matara	1991
92001	Warne	Shane	Australia	1969	Melbourne	1992
92002	Murlitharan	Muthiah	Sri Lanka	1972	Kandy	1992
92003	UI-Huq	Inzamam	Pakistan	1970	Multan	1992
93002	Streak	Heath	Zimbabwe	1974	Bulawayo	1993
93003	Kirsten	Gary	South Africa	1967	Cape Town	1993
94002	Vaas	Chaminda	Sri Lanka	1974	Mattumagala	1994
94004	Fleming	Stephen	New Zealand	1973	Christchurch	1994
95001	Ponting	Ricky	Australia	1974	Launceston	1995
95002	Pollock	Shaun	South Africa	1973	Port Elizabeth	1995
95003	Kallis	Jacques	South Africa	1975	Cape Town	1995
96001	Dravid	Rahul	India	1973	Indore	1996
96002	Gibbs	Herschelle	South Africa	1974	Cape Town	1996
96003	Ganguly	Saurav	India	1972	Calcutta	1996
97004	Vettori	Daniel	New Zealand	1979	Auckland	1997
98001	Afridi	Shahid	Pakistan	1980	Khyber Agency	1998
98002	Singh	Harbhajan	India	1980	Jalandhar	1998
99001	Lee	Breet	Australia	1976	Wollongong	1999
99002	Gilchrist	Adam	Australia	1971	Bellingen	1999
99003	Vaughan	Michale	England	1974	Manchester	1999

29 rows in set (0.0007 sec)

Q1c. Print Bowling table.

mysql> select * from Bowling;

MySQL localhost:3306 ssl matches_info_by_parth SQL > select* from bowling;

MatchID	PID	NOvers	Maidens	Nruns	Nwickets
2689	99001	10	0	58	1
2689	24001	3	0	27	1
2689	23001	3	0	15	0
2755	94002	9	1	40	1
2755	92002	10	0	56	1
2755	91001	4	0	29	0
2755	23001	10	0	53	2
2755	98002	10	0	40	3
2689	98002	10	0	44	1

9 rows in set (0.0006 sec)

Q1d. Print Batting table.

mysql> select * from Batting;

MySQL	localhost:3306 ssl	matches_info_by_parth	SQL	> select * from batting;					
MatchID	PID	POrder	HOut	FOW	NRuns	Mts	Nballs	Fours	Six
2755	23001	3	C	51	0	12	6	0	0
2755	25001	5	C	232	71	104	80	4	0
2755	91001	1	C	74	60	85	52	8	2
2755	94002	7	LBW	157	17	23	29	1	0
2755	92002	11	NO	NO	1	11	7	0	0
2689	89001	2	C	205	91	176	121	7	0
2689	23001	4	C	175	38	27	38	2	2
2689	25001	5	C	240	36	52	37	2	1
2689	99002	1	C	2	2	3	3	0	0
2689	95001	3	C	8	1	9	7	0	0
2689	24001	5	C	123	42	81	56	2	1
2689	99001	8	B	228	7	20	12	0	0
2689	27001	9	C	255	7	7	7	1	0
2755	27001	9	B	257	2	7	6	0	0
2689	98002	8	LBW	249	3	7	3	0	0
2755	98002	8	RO	253	2	7	4	0	0

16 rows in set (0.0007 sec)

Q2. Find the Ids, First name, Last name as Surname of all cricket players.

mysql> select PlayerID,Fname,Lname as Surname from Player;

```
mysql:localhost:3306 ssl matches_info_by_parth SQL> select PlayerID,Fname,Lname as Surname from Player;
```

PlayerID	Fname	Surname
21001	Virender	Sehwag
23001	Yuvraj	Singh
24001	Andrew	Symonds
25001	MS	Dhoni
27001	Praveen	Kumar
27002	Ishant	Sharma
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
91001	Sanath	Jayasuriya
92001	Shane	Warne
92002	Muthiah	Murthitharan
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94002	Chaminda	Vaas
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravid
96002	Herschelle	Gibbs
96003	Saurav	Ganguly
97004	Daniel	Vettori
98001	Shahid	Afridi
98002	Harbhajan	Singh
99001	Breet	Lee
99002	Adam	Gilchrist
99003	Michale	Vaughan

29 rows in set (0.0005 sec)

Q3a. Find all the information from table Player about players from India.

mysql> select * from Player where Country='India';

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > select * from player where country='India'
```

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
21001	Sehwag	Virender	India	1978	Delhi	2001
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007
89001	Tendulkar	Sachin	India	1973	Mumbai	1989
90002	Kumble	Anil	India	1970	Bangalore	1990
96001	Dravid	Rahul	India	1973	Indore	1996
96003	Ganguly	Saurav	India	1972	Calcutta	1996
98002	Singh	Harbhajan	India	1980	Jalandhar	1998

```
10 rows in set (0.0007 sec)
```

Q3b. Find the player Ids, First name & birth place of Indian cricket players.

```
mysql> select PlayerID,Fname,Bplace from Player where Country='India';
```

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > select PlayerID,fname,bplace from player where country ='India';
```

PlayerID	fname	bplace
21001	Virender	Delhi
23001	Yuvraj	Chandigarh
25001	MS	Ranchi
27001	Praveen	Meerut
27002	Ishant	Delhi
89001	Sachin	Mumbai
90002	Anil	Bangalore
96001	Rahul	Indore
96003	Saurav	Calcutta
98002	Harbhajan	Jalandhar

```
10 rows in set (0.0006 sec)
```

Q4. Find all the information from table Player about players from India who were born after 1980

```
mysql> select * from Player where Country='India' and Yborn > 1980;
```

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > select * from player where country='India' and Yborn>1980;
```

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007

```
4 rows in set (0.0017 sec)
```

Q5. Find the details of all matches played by Pakistan

```
mysql> SELECT * FROM ODIMatch WHERE Team1='Pakistan' OR Team2='Pakistan';
```

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > select * from odimatch where team1='Pakistan' or team2='Pakistan';
```

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2717	Pakistan	India	Karachi	2008-06-26	Team2

```
4 rows in set (0.0006 sec)
```

Q6a. Find the details of all matches played by India as a Second team.

```
mysql> SELECT * FROM ODIMatch WHERE Team2='India';
```

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > select * from odimatch where team2='India';
```

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2377	West Indies	India	Kingston	2006-05-18	Team2
2404	Sri Lanka	India	Colombo	2006-08-16	Abandoned
2449	South Africa	India	Cape Town	2006-11-26	Team1
2611	England	India	Southampton	2007-08-21	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2
2717	Pakistan	India	Karachi	2008-06-26	Team2
2750	Sri Lanka	India	Colombo	2008-08-24	Team2
2755	Sri Lanka	India	Colombo	2008-08-24	Team2

```
12 rows in set (0.0007 sec)
```

Q6b. Find the details of all matches played by India or Australia as Team1.

```
mysql> SELECT * FROM ODIMatch WHERE Team1='Australia' OR Team1='India';
```

```
mysql> SELECT * FROM ODIMatch WHERE Team1 IN ('Australia','India');
```

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT * FROM ODIMatch WHERE Team1 IN ('Australia','India');
```

MatchID	Team1	Team2	Ground	Date	Winner
2357	India	England	Delhi	2006-03-28	Team1
2440	India	Australia	Mohali	2006-10-29	Team2
2480	India	West Indies	Nagpur	2007-01-21	Team1
2493	India	West Indies	Vadodara	2007-01-31	Team1
2520	India	Sri Lanka	Rajkot	2007-02-11	Team2
2632	India	Australia	Mumbai	2007-10-17	Team1
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2681	India	Sri Lanka	Adelaide	2008-02-19	Team1
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2

```
11 rows in set (0.0010 sec)
```

Q6c. Find the match details Team2, Winner & ground in which India won the toss

(assume Team1 is toss winner)

mysql> SELECT Team2, Winner, Ground FROM ODIMatch WHERE Team1 = 'India';

```
MySQL localhost:3306 ssl matches_info_by_parth SQL> select team2, winner, ground from odimatch where team1='India';
```

team2	winner	ground
England	Team1	Delhi
Australia	Team2	Mohali
West Indies	Team1	Nagpur
West Indies	Team1	Vadodara
Sri Lanka	Team2	Rajkot
Australia	Team1	Mumbai
Pakistan	Team1	Guwahati
Sri Lanka	Team1	Adelaide

8 rows in set (0.0007 sec)

Q7. Find Ids of all players that have bowled in an ODI match in the database

mysql> SELECT DISTINCT PID FROM Bowling;

```
MySQL localhost:3306 ssl matches_info_by_parth SQL> SELECT DISTINCT PID FROM Bowling;
```

PID
23001
24001
91001
92002
94002
98002
99001

7 rows in set (0.0006 sec)

Q8. Find names of teams and grounds where India has played an ODI match outside India.

(assume Team1 is the host country)

mysql> SELECT DISTINCT Team1, Ground FROM ODIMatch WHERE Team2='India';

```
MySQL localhost:3306 ssl matches_info_by_parth SQL> SELECT DISTINCT Team1, Ground FROM ODIMatch WHERE Team2='India';
```

Team1	Ground
Pakistan	Peshwar
Pakistan	Rawalpindi
West Indies	Kingston
Sri Lanka	Colombo
South Africa	Cape Town
England	Southampton
Australia	Melbourne
Australia	Sydney
Australia	Brisbane
Pakistan	Karachi

10 rows in set (0.0019 sec)

Q9. Find Ids of all players that batted in match 2755

mysql> SELECT PID FROM Batting WHERE MatchID=2755;

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT PID FROM Batting WHERE MatchID=2755;
+-----+
| PID |
+-----+
| 23001 |
| 25001 |
| 91001 |
| 94002 |
| 92002 |
| 27001 |
| 98002 |
+-----+
7 rows in set (0.0007 sec)
```

Q10a. Display list of ground names where Australia has played as Team1

mysql> SELECT Ground FROM ODIMatch WHERE Team1='Australia';

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT Ground FROM ODIMatch WHERE Team1='Australia';
+-----+
| Ground |
+-----+
| Melbourne |
| Sydney |
| Brisbane |
+-----+
3 rows in set (0.0006 sec)
```

Q10b. Display sorted list of ground names where Australia has played as Team1

mysql> SELECT Ground FROM ODIMatch WHERE Team1='Australia' ORDER BY Ground;

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT Ground FROM ODIMatch WHERE Team1='Australia' ORDER BY Ground;
+-----+
| Ground |
+-----+
| Brisbane |
| Melbourne |
| Sydney |
+-----+
3 rows in set (0.0021 sec)
```

Q11. Find name of all players whose last name starts with Sin

mysql> SELECT Fname,Lname FROM Player WHERE Lname LIKE 'Sin%';

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT Fname,Lname FROM Player WHERE Lname LIKE 'Sin%';
+-----+-----+
| Fname | Lname |
+-----+-----+
| Yuvraj | Singh |
| Harbhajan | Singh |
+-----+-----+
2 rows in set (0.0019 sec)
```

Q12. Find Ids, First name of all players who played in match 2755.

mysql> SELECT PlayerID,Fname FROM Player WHERE PlayerID IN (SELECT PID FROM Batting WHERE MatchID=2755);

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT PlayerID,Fname FROM Player WHERE PlayerID IN (SELECT PID FROM Batting WHERE MatchID=2755);
```

PlayerID	Fname
23001	Yuvraj
25001	MS
91001	Sanath
94002	Chaminda
92002	Muthiah
27001	Praveen
98002	Harbhajan

7 rows in set (0.0010 sec)

Q13. Find match Ids of all matches in which Tendulkar batted.

mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname="Tendulkar");

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');
```

MatchID
2689

1 row in set (0.0006 sec)

Q14. Find match information of all matches in which dhoni has batted.

mysql> SELECT MatchID,Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN(SELECT PlayerID FROM Player WHERE Lname='Dhoni'));

```
MySQL localhost:3306 ssl matches_info_by_parth SQL > SELECT MatchID,Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN(SELECT PlayerID FROM Player WHERE Lname='Dhoni'));
```

MatchID	Team1	Team2	Ground	Date
2755	Sri Lanka	India	Colombo	2008-08-24
2689	Australia	India	Brisbane	2008-03-04

2 rows in set (0.0008 sec)